



Habib University

EE-426/CE-442 Cryptography and Network Security - Spring 2026

Instructor: Farhan Khan

Time = 40 minutes

Quiz 02

Max Points: 40

Instructions:

- i. **Smart watches and similar electronics are strictly NOT allowed.**
- ii. **Answer sheets should contain all steps, working, explanations, and assumptions.**
- iii. Single-sided A4-sized cheat sheet is allowed.
- iv. Attempt the quiz on clean papers with black/blue ink.
- v. Print your name and HU ID on all sheets.
- vi. Turn in your question paper along with your answer sheet(s).
- vii. You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

CLO Assessment:

This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	Apply the concepts from number theory and algebraic structures in understanding the design of cryptographic algorithms.	
CLO 2	Evaluate the suitability of message integrity, message authentication, and key management methods when applying in a given security scenario.	✓
CLO 3	Explain the setup, algorithms, and security issues in existing symmetric-key and asymmetric-key cryptosystems.	
CLO 4	Apply cryptography-based network security technologies in the design of networked information systems.	

Question 1 [10 points]:

- Define what is meant by a cryptographic hash function. [3 points]
- Describe the idea of the Merkle-Damgård scheme with the help of a diagram and discuss why this idea is so important for the design of a cryptographic hash function. [3 points]
- Briefly discuss the two groups of compression functions used in Merkle-Damgård scheme? [2 points]
- List two families of hash functions that belong to the first group. [1 point]
- List a scheme that belongs to the second group. [1 point]

Answer:

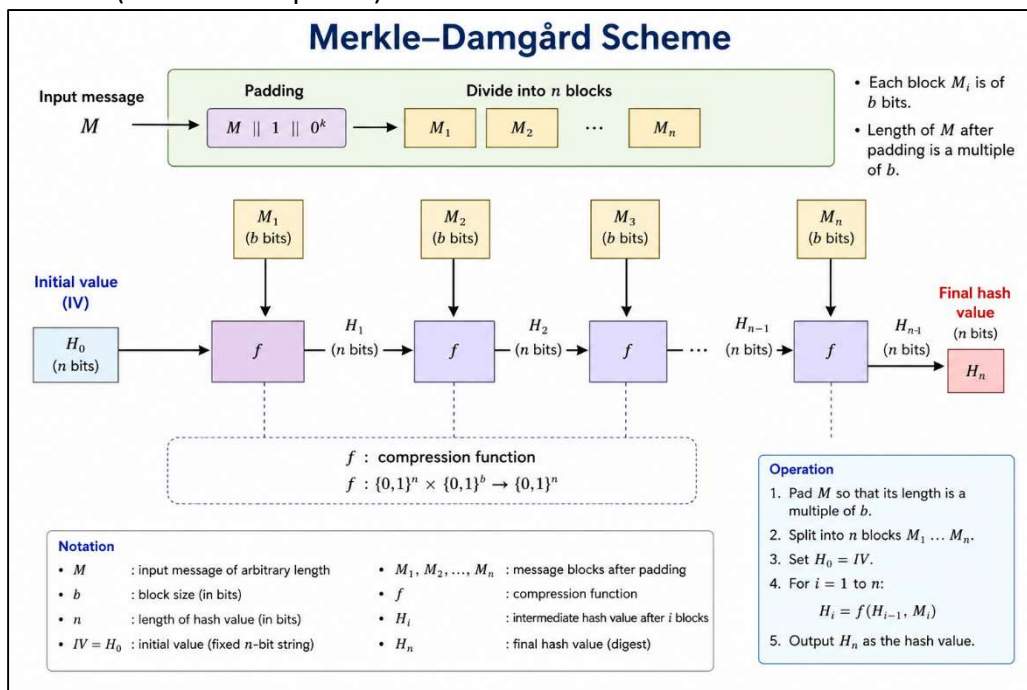
a) A cryptographic hash function is a deterministic algorithm that takes an input of arbitrary length and produces a fixed-length output (hash/digest), such that it is:

- Preimage resistant (hard to find input from hash)
- Second preimage resistant (hard to find another input with same hash)
- Collision resistant (hard to find two different inputs with same hash)

b) It is a construction method to build a secure hash function from a fixed-length compression function by processing input in blocks iteratively.

Importance:

- Allows hashing of arbitrary-length messages
- Converts a secure compression function $\rightarrow$ full hash function
- Foundation of major hash functions like MD5, SHA-1, SHA-256
- Security property: If compression function is collision resistant $\rightarrow$ hash is collision resistant (under assumptions)



c) Two Groups of Compression Functions in Merkle–Damgård

1. Dedicated hash compression functions
 - Specifically designed for hashing from scratch (not derived from block ciphers)
2. Block-cipher-based compression functions
 - Built using symmetric encryption algorithms

d) MD5 and SHA-2

e) WHIRLPOOL

Question 2 [5 points]:

In SHA512 hashing algorithm, the majority function is given by:

$$(A_i \text{ AND } B_i) \oplus (B_i \text{ AND } C_i) \oplus (C_i \text{ AND } A_i)$$

Suppose we apply the majority function on A, B, and C buffers. If the leftmost hexadecimal digits of these buffers are 0x7, 0xA, and 0xC respectively, what is the leftmost digit of the result?

Answer:

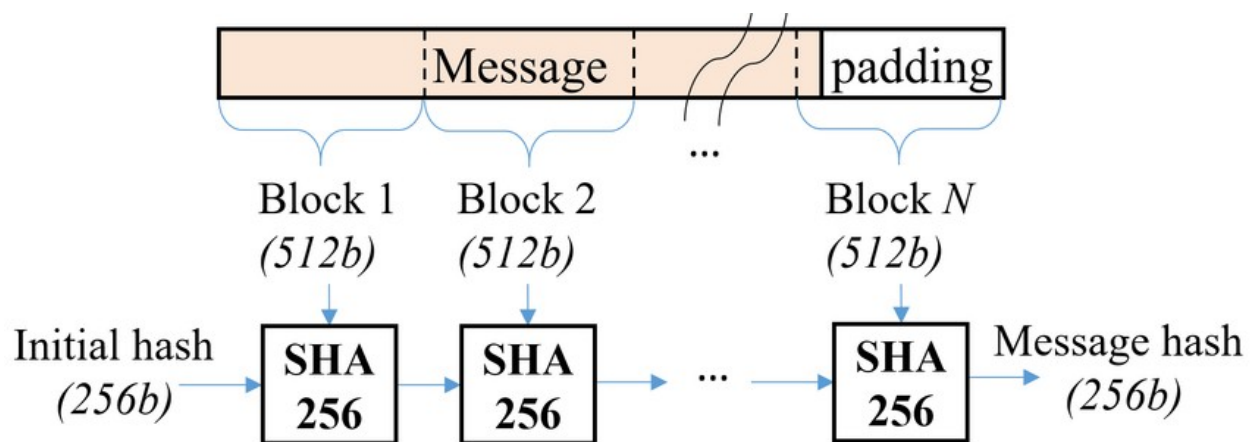
The digits in binary are 0111, 1010, and 1100.

- The first bits are 0, 1, and 1. The result is 1.
- The second bits are 1, 0, and 1. The result is 1.
- The third bits are 1, 1, and 0. The result is 1.
- The fourth bits are 1, 0, and 0. The result is 0.

The result is 1110, or 0xE in hexadecimal.

Question 3 [5 points]:

The following Figure shows the general structure of popular cryptographic hash function.



- i. Find out the size of the message block used in this algorithm & the size of the message digest which is being computed in this algorithm.

- ii. If there is a requirement to use a hashing algorithm with the best possible collision resistance, would you recommend this algorithm. If not, what will be your recommendation?

Answer:

- i. 512-bit block size and 256-bit message digest
- ii. SHA256 produces a 256-bit output, and SHA512 produces a 512-bit output. Generally, the longer the output, the more secure the hash function, as it reduces the chances of collisions (two different inputs producing the same output hash). However, the longer the output, the more computational resources and time required to generate and process the hash. Therefore, there is a trade-off between security and performance. SHA256 overall is a good compromise between security and performance, as it offers a high level of resistance to attacks and collisions, while being relatively fast and easy to implement but SHA512 is a better option than SHA256 in terms of collision resistance.

Question 4 [10 points]:

Using the RSA scheme, let $p = 809$, $q = 751$, and $d = 23$. Calculate the public key e . Then:

- i. Sign and verify a message with $M_1 = 5$. Call the signature S_1 .
- ii. Sign and verify a message with $M_2 = 10$. Call the signature S_2 .

Answer:

Values of p and q are given in the question,

$$n = p \times q = 809 \times 751 = 607559$$

$$\phi(n) = (p-1) \times (q-1) = (809-1) \times (751-1) = 606000.$$

Since $d = 23$, we have $e = d^{-1} \bmod \phi(n) = 158087$

(i)

$$S_1 = M_1^d \bmod n = 5^{23} \bmod 607559 = 186465$$

$$M_1 = S_1^e \bmod n = 186465^{158087} \bmod 607559 = 5$$

(ii)

$$S_2 = M_2^d \bmod n = 10^{23} \bmod 607559 = 488214$$

$$M_2 = S_2^e \bmod n = 488214^{158087} \bmod 607559 = 10$$

Question 5 [10 points]:

Using the ElGamal scheme, let $p = 881$ and $d = 700$. Public key consists of values of p , e_1 and e_2 .

Choose $r = 17$. Then:

- i. Find the values of e_1 and e_2 .
- ii. Find the value of S_1 and S_2 if $M = 40$.

Answer:

Values of p and d are given in the question,

$p = 881$ $d = 700$.

(i)

We choose $e_1 = 3$ (primitive root of 881).

Then $e_2 = e_1^d \bmod p = 471$.

$r = 17$

(ii)

$S_1 = e_1^r \bmod p = 3^{17} \bmod 881 = \mathbf{540}$

$S_2 = (M - d \times S_1) r^{-1} \bmod (p-1) = (40 - 700 \times 540) 17^{-1} \bmod 880 = (40 - 700 \times 540) 17^{-1} \bmod 880 = 440 \times 673 \bmod 880 = \mathbf{440}$

where $17^{-1} \bmod 880 = 673$